# Eliminating the Observer Effect: Wave Function Collapse as Deterministic Topological Reconnection in a Condensate Vacuum

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**Repository:** https://github.com/infosave2007/vmf

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## Abstract

The measurement problem and the effective localisation of quantum outcomes remain among the most contentious issues in the foundations of quantum mechanics. In contemporary discussions, the problem is usually formulated not in terms of a conscious observer, but in terms of interaction, decoherence, interpretational choice, and possible objective-collapse dynamics. This paper develops an NVG-inspired ontological model in which the quantum vacuum is treated as a macroscopic, non-perturbative Quantum Chromodynamics (QCD) condensate. In this picture, an elementary particle is represented as a localized topological defect propagating together with an extended strain-like excitation of the vacuum medium. Wave-function collapse is then reinterpreted as deterministic topological reconnection triggered by a sufficiently strong environmental inhomogeneity, such as a detector lattice. The scope of the paper is precise: it formulates the model, writes a testable phenomenological ansatz, and derives first benchmark constraints on its collapse-rate parameter space. It does not yet derive the Born rule, repeated-measurement statistics, Bell/EPR correlations, or precision bounds on nonlinear collapse-like effects; those are the next derivational layer of the programme.

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# 1. Introduction

Since the formalization of quantum mechanics in the 1920s, the interpretation of the wave function and the status of measurement have remained unsettled. In modern foundations, the central issue is usually not whether a conscious observer creates reality, but how unitary evolution, decoherence, definite outcomes, and observed measurement statistics fit together in a single coherent framework. The standard formalism works exceptionally well operationally, yet questions remain about what exactly the wave function represents and why measurements yield localized outcomes.

This tension produces the familiar interpretational landscape: Copenhagen-style instrumental readings, decoherence-based accounts, many-worlds, de Broglie-

Bohm pilot-wave theory, and spontaneous objective-collapse models such as GRW or Penrose-type proposals. Each resolves part of the conceptual difficulty while leaving open questions of ontology, dynamics, or empirical distinctiveness.

In this letter, we introduce a different approach using the Null-Vector Gravity (NVG) framework. By upgrading the ontological status of the vacuum to a structured physical medium anchored by the QCD mass gap, we reinterpret the quantum state as a real strain-like excitation of the vacuum fabric. The goal of the article is direct: to formulate localisation as an effective dynamical reconnection process in such a medium and to show that this formulation already leads to a concrete phenomenological ansatz with nontrivial parameter-space constraints.

## 2. The Vacuum as a Macroscopic Condensate

The core premise of the NVG framework is that the mass of fundamental particles is an effective coupling to a ubiquitous vacuum field, scaling with the non-perturbative QCD anchor $M_{\Omega,0} = 859$ MeV. The vacuum is a dense, highly structured condensate.

In this framework, an elementary particle is not an infinitely small point metric. It is a stable topological defect—a localized "knot" or vortex—within the continuous vector field of the vacuum. Because the vacuum is an elastic medium, the energetic presence of this defect induces a strain that extends indefinitely outward.

Therefore, a traveling electron is a physical soliton. The core of the defect carries the quantized charge and the bulk of the rest mass, while the surrounding "wake" in the vacuum fluid constitutes the spatially extended wave.

Historically, scalar 3D solitons have faced Derrick's theorem (1964), which states that static localized solitons are unstable in 3+1 dimensions. However, this restriction is bypassed when we treat the particle as a *dynamic resonance mode (standing wave packet)* rather than a static configuration, as shown by White et al. (2026). In this picture, quadratic dispersion ($\omega = Dq^2$) in a compressible vacuum allows stable, oscillatory localized wave resonances to persist.

## 3. The Double-Slit Experiment Reinterpreted

When a single electron is fired towards a double-slit apparatus, the Copenhagen interpretation asserts that the point-particle enters a superposition of states, traversing both slits simultaneously as a probability wave, interfering with itself, and only deciding where to land upon striking the detector.

In the NVG fluid-dynamic model, the process is described in effectively classical and deterministic language at the level of ontology. The electron is treated as a physical wave of vacuum displacement. As this spatially extended wave packet approaches the barrier, the broad "wake" physically passes through both slits simultaneously, creating two secondary wavefronts of vacuum strain that undergo interference, exactly analogous to ripples in a fluid surface.

The core topological defect (the localized knot of mass-energy) passes through only one of the slits, guided by the local gradients of the total interference pattern generated by its own extended wake. The electron is "riding" its own self-generated standing wave in the vacuum condensate.

# 4. Deterministic Topological Reconnection (Collapse)

The defining moment of quantum mechanics is the registration of the particle on the detector screen as a single, highly localized point. Why does the extended interference wave "collapse"?

In the NVG framework, the detector screen is a dense macroscopic lattice of other stable topological defects (atoms). When the diffuse, extended wavefront of the electron's vacuum strain encounters this dense barrier, it experiences a drastic impedance mismatch. The continuous conformal field lines of the vacuum strain are violently interrupted.

This triggers a well-known phenomenon in fluid dynamics and plasma physics: **Topological Reconnection**. The vacuum vector field must rapidly reconfigure to minimize energy. The non-linear tension in the vacuum fabric causes the extended wave envelope to rapidly contract, "snapping" the delocalized strain energy back toward the core topological defect.

The wave does not vanish probabilistically; in this picture it retracts physically into a localized point-knot. At the level of the present note, however, this remains a qualitative dynamical proposal rather than a complete statistical theory: we describe why localisation might occur in a structured medium, but we do not yet derive the full probability law for the distribution of outcomes.

# 5. Scope and Open Problems

For this proposal to function as a serious physical alternative rather than only an ontological sketch, at least four missing elements must still be supplied.

1. **Unitary evolution and the Born rule.** The present article does not derive the standard linear Schrödinger evolution or the Born rule from first principles. However, the emergence of a linear Schrödinger-like equation from the underlying medium can be modeled via the Madelung hydrodynamic formulation (White et al., 2026). By mapping the Goldstone phase $\theta$ of the $U(1)$ vacuum condensate to a velocity potential ($S = \hbar\theta$), spatial variations in the condensate density $\mathcal{W}(x)$ yield the Madelung quantum

potential $Q(x) = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\mathcal{W}}}{\sqrt{\mathcal{W}}}$. A local depression (melting) in the vacuum field density generates a physical self-localization well, providing a mathematical link between classical $\mathcal{W}$-dynamics and the Schrödinger equation.

2. **Repeated measurements and detector statistics.** A viable theory must explain not only a single localisation event, but also the observed ensemble frequencies across many runs, including the stability of standard quantum statistics.
3. **EPR/Bell consistency.** Any deterministic vacuum-medium account must state explicitly how it reproduces nonlocal quantum correlations without contradicting Bell-test data.
4. **Experimental distinctiveness.** The model should produce at least one clean, quantitative deviation from standard quantum mechanics or from known objective-collapse models, together with a scale at which that deviation becomes measurable.

These omissions are not secondary details. They define the current boundary between an interesting conceptual picture and a complete physical theory. The present note should therefore be read as a heuristic reformulation of localisation, not yet as a finished alternative to standard quantum mechanics.

# 6. Minimal Phenomenological Requirements

Even at the present exploratory stage, the proposal can be made more physical by stating what kinds of observable consequences would follow if a vacuum-medium reconnection mechanism were real. The point is not that the current note has derived these effects quantitatively, but that any successful completion of the model must eventually land in this space of checks.

1. **Density- and environment-dependent localisation scale.** If collapse-like localisation is triggered by a strong impedance mismatch between an extended vacuum excitation and a dense environment, then the localisation rate should depend on material properties of the detector rather than only on an abstract measurement postulate. In a mature version of the theory, this would imply a calculable dependence on lattice density, dielectric structure, or effective coupling to the vacuum medium.
2. **Mass and size dependence similar to objective-collapse models, but with a different scaling law.** GRW/CSL-type models are constrained because they predict a collapse rate that grows with system size or mass. A vacuum-reconnection model would likewise need a concrete scaling law for mesoscopic objects, molecules, nanospheres, and mechanical resonators. Without such a law, the proposal cannot be confronted with matter-wave interferometry.
3. **Null limit in ordinary microscopic regimes.** The model must reproduce the observed success of standard quantum mechanics in atomic, optical, and low-energy scattering experiments. This means that any effective nonlinearity or reconnection threshold has to be parametrically tiny

for isolated microscopic systems, otherwise the theory would already be excluded by precision interference measurements.

4. **Compatibility with heating and spontaneous-radiation bounds.** Known collapse-like models are strongly constrained by anomalous heating, spontaneous photon emission, and loss of coherence in optomechanical or cold-atom systems. Any nonlinear vacuum-medium completion must either avoid these effects or predict them below current experimental sensitivity.

These points define the minimum phenomenological bar. A useful next paper would not merely repeat the ontology, but derive a collapse rate

$$\Gamma_{\text{coll}} = \Gamma_{\text{coll}}(m, L, \rho_{\text{env}}, \nabla\rho, \text{coupling parameters}),$$

or an equivalent measurable quantity that can be placed against existing interferometry and decoherence data.

# 7. A Minimal Phenomenological Collapse Ansatz

Even without a full microscopic derivation, it is useful to write the simplest working parametrisation that captures the physical intuition of the model. In the NVG picture, localisation should become relevant only when three ingredients are present simultaneously: a sufficiently extended excitation, a sufficiently strong environmental density gradient, and a coupling of the excitation to the vacuum medium. The most conservative ansatz is therefore a thresholded collapse rate of the form

$$\Gamma_{\text{coll}}^{(\text{NVG})} = \Gamma_0 \left(\frac{m}{m_0}\right)^{\alpha} \left(\frac{L}{L_0}\right)^{\beta} \left(\frac{\rho_{\text{env}}}{\rho_0}\right)^{\gamma} \left(\frac{|\nabla\rho|}{|\nabla\rho|_0}\right)^{\delta} \Theta(L - L_{\text{crit}}),$$

where $m$ is the system mass, $L$ its spatial extent or coherence length, $\rho_{\text{env}}$ and $|\nabla\rho|$ characterize the detector environment, $\Gamma_0$ is a reference rate, and $\Theta$ is a threshold factor expressing that microscopic systems below some critical scale should remain effectively quantum.

The meaning of this ansatz is limited but useful.

1. If $\alpha > 0$ and $\beta > 0$, larger and more extended objects localise faster, qualitatively like GRW/CSL.
2. If $\gamma > 0$ or $\delta > 0$, localisation is not purely intrinsic to the wave function but depends on the detector medium, which is the distinctive feature of the NVG picture.
3. If $L < L_{\text{crit}}$ or the environmental gradient is weak, then $\Gamma_{\text{coll}}^{(\text{NVG})} \to 0$ and standard quantum interference should be recovered.

This formula is not yet a derivation. It is a placeholder that turns the proposal into something that can be compared with experiment. The immediate job of a follow-up paper would be to determine whether any parameter region

$$(\Gamma_0, \alpha, \beta, \gamma, \delta, L_{\text{crit}})$$

survives existing matter-wave, optomechanical, and spontaneous-heating limits while still producing effective localisation in macroscopic detectors.

The main advantage of writing the ansatz explicitly is that it clarifies what makes the model different from standard CSL-like collapse: not merely growth with mass, but an explicit dependence on environmental density and density gradients. That difference is exactly what can make the theory testable.

## 7.1. Illustrative Parameter Corridor

At the present stage, the least committal benchmark is to keep the scaling exponents minimal and positive,

$$\alpha \approx 1, \qquad \beta \approx 1, \qquad \gamma \approx 1,$$

and then treat $\delta$ and $L_{\text{crit}}$ as the genuinely distinguishing NVG parameters. In words: collapse strengthens roughly linearly with system size, mass, and environmental density, while detector inhomogeneity enters through the density-gradient exponent $\delta$.

For a first phenomenological scan, two benchmark regimes are the most useful.

1. **Soft medium-sensitive benchmark:**

   $$(\alpha, \beta, \gamma, \delta) = (1, 1, 1, 1).$$

   This is the smallest deformation away from standard decoherence-like intuition. It asks only whether adding explicit environmental dependence already improves the ontology without immediately violating interference bounds.
2. **Gradient-dominated benchmark:**

   $$(\alpha, \beta, \gamma, \delta) = (1, 1, 1, 2).$$

   This version encodes the stronger NVG intuition that localisation is triggered primarily by sharp detector inhomogeneity rather than by ambient density alone.

The threshold scale $L_{\text{crit}}$ should be chosen so that known microscopic interference remains intact. A conservative first-pass requirement is therefore that $L_{\text{crit}}$ lie above atomic and small-molecular coherence scales but below obviously macroscopic detector scales,

$$10^{-8}\ \text{m} \lesssim L_{\text{crit}} \lesssim 10^{-6}\ \text{m},$$

to be refined later against real interferometry data. This is not a derived range; it is only a phenomenological corridor that keeps the model from being trivially excluded at one end or operationally useless at the other.

Finally, $\Gamma_0$ must satisfy two opposite inequalities:

1. For isolated microscopic systems,

$$\Gamma_{\mathrm{coll}}^{(\mathrm{NVG})} t_{\mathrm{exp}} \ll 1,$$

   so that standard interference survives over the experimental coherence time $t_{\mathrm{exp}}$.
2. For a macroscopic detector environment,

$$\Gamma_{\mathrm{coll}}^{(\mathrm{NVG})} t_{\mathrm{det}} \gg 1,$$

   so that localisation is effectively complete within the detector response time $t_{\mathrm{det}}$.

These inequalities already define a concrete parameter-fitting problem, even before a first-principles derivation exists.

## 7.2. Comparison with Standard QM and GRW/CSL

The value of the NVG ansatz is easier to see when compared with the two main alternatives it would have to compete with.

| Feature | Standard QM + Decoherence | GRW / CSL | NVG Reconnection Ansatz |
|---|---|---|---|
| Basic ontology | Wave function plus measurement postulate or interpretation | Modified stochastic collapse dynamics | Real vacuum medium with topological defects and reconnection |
| Microscopic evolution | Linear and unitary | Nonlinear / stochastic modification | Intended to reduce to linear QM below threshold |
| Why outcomes localize | Decoherence explains suppression of interference but not unique outcome by itself | Objective stochastic collapse | Effective reconnection in a dense structured environment |
| Environment dependence | Strong through decoherence, but collapse not fundamental | Usually weak or absent in the fundamental collapse term | Explicit dependence on density and density gradients |

| Feature | Standard QM + Decoherence | GRW / CSL | NVG Reconnection Ansatz |
|---|---|---|---|
| Key control parameters | Hamiltonian and environmental coupling | Collapse rate and localization length | $\Gamma_0$, exponents $(\alpha, \beta, \gamma, \delta)$, and $L_{\text{crit}}$ |
| Born rule status | Built into formalism | Built into stochastic law | Not yet derived |
| Main experimental tests | Interferometry, decoherence control, Bell tests | Heating, spontaneous radiation, interferometry, optomechanics | Same tests, plus detector-material and density-gradient dependence |

This comparison makes the strategic point of the article more explicit. The NVG proposal is not competitive with standard QM or GRW/CSL because it is already more complete; it is interesting only if its explicit medium dependence leads to a quantitatively distinct and experimentally testable scaling law.

## 7.3. First Worked Order-of-Magnitude Estimate

The ansatz is simple enough that one can already extract a crude self-consistency corridor. The goal of this exercise is not a rigorous bound, but to check whether there exists any nonempty parameter range in which the model simultaneously preserves observed microscopic interference and still allows rapid localisation in a macroscopic detector.

Take the soft benchmark

$$(\alpha, \beta, \gamma, \delta) = (1, 1, 1, 1),$$

with reference scales

$$m_0 = 1 \text{ amu}, \qquad L_0 = 1 \text{ nm}, \qquad \rho_0 = 10^3 \text{ kg m}^{-3}, \qquad |\nabla\rho|_0 = 10^{13} \text{ kg m}^{-4},$$

where the density-gradient scale corresponds roughly to a condensed-matter interface over an atomic length.

For a representative large-molecule interferometry benchmark, we take

$$m_{\text{int}} \sim 2.5\times10^4 \text{ amu}, \qquad L_{\text{int}} \sim 10^{-7} \text{ m}, \qquad \rho_{\text{int}} \sim 10^{-13} \text{ kg m}^{-3}, \qquad |\nabla\rho|_{\text{int}} \sim 10^{-13} \text{ kg m}^{-4},$$

with an experimental coherence time of order

$$t_{\text{exp}} \sim 10^{-2} \text{ s}.$$

These values are chosen to be representative of high-vacuum matter-wave interference with large molecules in the Vienna class of experiments reaching masses beyond $2.5 \times 10^4$ amu.

The dimensionless prefactor in the NVG rate is then

$$\left(\frac{m_{\rm int}}{m_0}\right)\left(\frac{L_{\rm int}}{L_0}\right)\left(\frac{\rho_{\rm int}}{\rho_0}\right)\left(\frac{|\nabla\rho|_{\rm int}}{|\nabla\rho|_0}\right) \sim 2.5\times 10^{-36},$$

so preserving interference requires

$$\Gamma^{(\rm NVG)}_{\rm coll} t_{\rm exp} \ll 1 \quad \Longrightarrow \quad \Gamma_0 \ll 4\times 10^{37}\ {\rm s}^{-1}.$$

Now compare this with a detector benchmark: a micron-scale active region,

$$m_{\rm det}\sim 10^{12}\ {\rm amu}, \qquad L_{\rm det}\sim 10^{-6}\ {\rm m}, \qquad \rho_{\rm det}\sim 10^{3}\ {\rm kg\,m^{-3}}, \qquad |\nabla\rho|_{\rm det}\sim 10^{13}\ {\rm kg\,m^{-4}},$$

and a response time

$$t_{\rm det}\sim 10^{-9}\ {\rm s}.$$

The corresponding prefactor is

$$\left(\frac{m_{\rm det}}{m_0}\right)\left(\frac{L_{\rm det}}{L_0}\right)\left(\frac{\rho_{\rm det}}{\rho_0}\right)\left(\frac{|\nabla\rho|_{\rm det}}{|\nabla\rho|_0}\right) \sim 10^{15},$$

so effective detector localisation requires

$$\Gamma^{(\rm NVG)}_{\rm coll} t_{\rm det} \gg 1 \quad \Longrightarrow \quad \Gamma_0 \gg 10^{-6}\ {\rm s}^{-1}.$$

Thus even this crude benchmark already yields a nonempty corridor,

$$10^{-6}\ {\rm s}^{-1} \ \ll\ \Gamma_0 \ \ll\ 4\times 10^{37}\ {\rm s}^{-1},$$

for the soft medium-sensitive ansatz. For the gradient-dominated benchmark with $\delta = 2$, the detector lower bound is unchanged while the interferometry upper bound is weakened even further, to roughly

$$\Gamma_0 \ll 4\times 10^{63}\ {\rm s}^{-1},$$

because the vacuum gradient suppression becomes extremely strong.

The lesson is not that the model is confirmed. The lesson is narrower: environmental suppression is so strong in high-vacuum interferometry that the ansatz is not immediately self-contradictory. This is encouraging, but it also means that the first serious constraints will likely come not from free-flight molecular interference alone, but from experiments that combine mesoscopic superpositions with well-characterized material environments, heating bounds, or detector-coupled decoherence measurements.

Figure 1 visualizes the same conclusion in the $(L_{\rm crit}, \Gamma_0)$ plane. The soft benchmark and the gradient-dominated benchmark both exhibit a broad nonempty corridor, but for different reasons: in the soft case the corridor is bounded above by interferometry only when $L_{\rm crit} < L_{\rm int}$, while in the gradient-dominated case

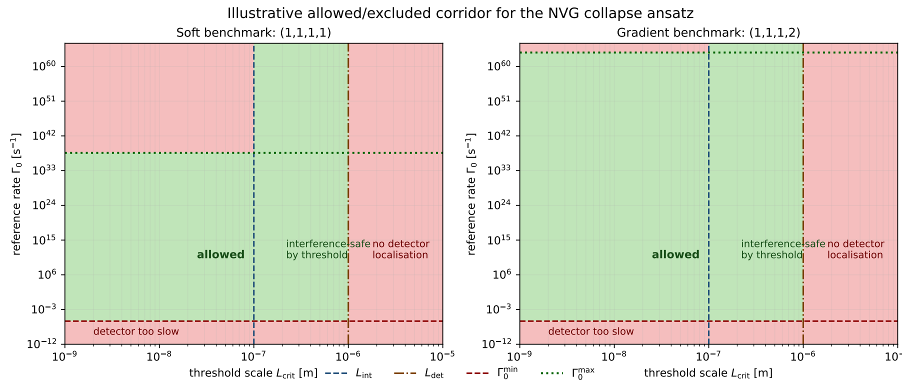


Figure 1: Illustrative allowed and excluded regions in the $(L_{\text{crit}}, \Gamma_0)$ plane for the two benchmark NVG collapse ansaetze. Green indicates benchmark-compatible regions; red indicates regions excluded either because microscopic interferometry would be spoiled or because detector localisation would be too slow.

vacuum suppression makes the upper bound effectively irrelevant on the plotted scale.

Figure 2 makes the same point in a more experiment-facing way by overlaying representative benchmark lines for atomic interferometry, $C_{60}$-class molecules, 25 kDa molecule interference, and a mesoscopic levitated-object target scale. These are not a full recast of the experimental literature; they are illustrative reference lines showing how the environmental suppression built into the NVG ansatz weakens vacuum interferometry bounds unless the threshold scale $L_{\text{crit}}$ lies below the physical coherence size of the system.

# 8. Concrete Tests That Would Matter

If the model is to move beyond conceptual reinterpretation, the following tests would be decisive.

1. **Matter-wave interferometry of large molecules or nanoparticles.** The theory should predict whether interference visibility begins to drop above a characteristic mass, size, or environmental-density threshold distinct from standard decoherence.
2. **Detector-material dependence of localisation.** If collapse is genuinely tied to topological reconnection in a structured medium, then localisation should depend weakly but measurably on detector composition or density profile, not only on the abstract presence of a measurement.
3. **Optomechanical and levitated-system bounds.** Precision experiments already place strong limits on collapse-like nonlinearities. Any viable NVG-inspired completion must map onto those limits and demon-

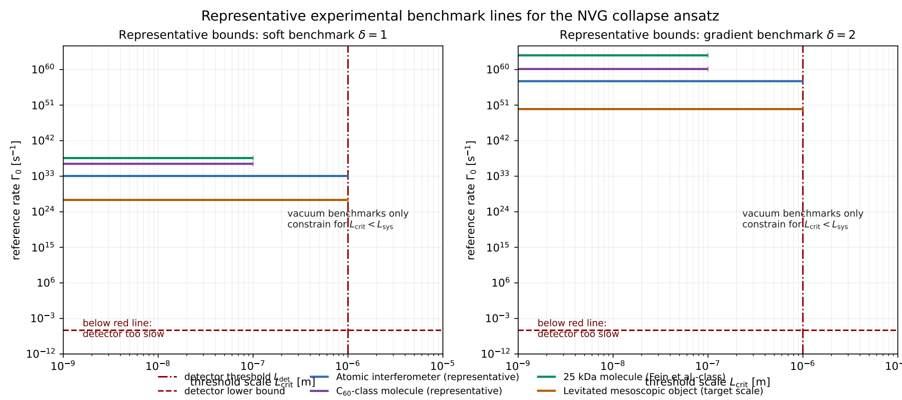


Figure 2: Representative benchmark lines in the $(L_{\text{crit}}, \Gamma_0)$ plane for atomic interferometry, molecule interference, and mesoscopic superposition targets. The red detector line is a lower bound required for fast localisation in a macroscopic detector, while colored horizontal lines show approximate upper bounds from preserving interference in representative vacuum experiments.

strate where its parameter space survives.

4. **Bell/EPR scenarios.** The theory must either reproduce standard nonlocal quantum correlations exactly or predict a deviation small enough to remain compatible with loophole-free Bell tests. This is not optional: it is one of the first places any deterministic medium theory will fail if it is not formulated carefully.

At present the note does not supply these calculations. It only identifies where they must appear for the proposal to become falsifiable in a serious experimental sense.

# 9. Conclusion

The Null-Vector Gravity framework provides a physical ontology for quantum processes in which localisation is interpreted as a medium-driven reconnection event rather than as a primitive measurement postulate. The article therefore replaces observer-centered language with an explicit medium-based dynamical picture and pushes that picture beyond pure ontology into a first phenomenological parameterisation.

1. **Wave and particle aspects are unified geometrically:** particles are modeled as topological defects coupled to an extended strain field in the vacuum medium.
2. **The wave function is reinterpreted ontologically:** the article treats it as a real excitation of the vacuum rather than only as an abstract bookkeeping device.

3. **Collapse is modeled qualitatively as reconnection:** measurement is re-described as a physical interaction between an extended excitation and a dense detector environment.

What the article does **not** yet show is equally important: it does not derive Born-rule probabilities, does not analyse Bell/EPR scenarios, does not establish compatibility with the modern decoherence framework, and does not confront the proposal with precision limits on nonlinear collapse-like dynamics. Those are the outstanding hard problems. The correct reading of the present paper is therefore sharp rather than apologetic: it is a deterministic NVG model of localisation with a concrete phenomenological ansatz and first benchmark constraints, but it is not yet a complete statistical completion of quantum theory and it is not yet a finished solution to the full measurement problem.

# Data Availability

Interactive 3D visualizations demonstrating the continuous propagation and deterministic topological reconnection of the NVG wave-particle mechanism are available in the public repository at: [https://github.com/infosave2007/vmf].

# References


1. Bell, J. S. "On the Einstein Podolsky Rosen paradox." *Physics Physique Fizika* 1, 195 (1964).
2. Ghirardi, G. C., Rimini, A., & Weber, T. "Unified dynamics for microscopic and macroscopic systems." *Physical Review D* 34, 470 (1986).
3. Bohm, D. "A Suggested Interpretation of the Quantum Theory in Terms of 'Hidden' Variables." *Physical Review* 85, 166 (1952).
4. Penrose, R. "On Gravity's role in Quantum State Reduction." *General Relativity and Gravitation* 28, 581 (1996).
5. Couder, Y., & Fort, E. "Single-Particle Diffraction and Interference at a Macroscopic Scale." *Physical Review Letters* 97, 154101 (2006).
6. Zurek, W. H. "Decoherence, einselection, and the quantum origins of the classical." *Reviews of Modern Physics* 75, 715 (2003).
7. Bassi, A., Lochan, K., Satin, S., Singh, T. P., & Ulbricht, H. "Models of wave-function collapse, underlying theories, and experimental tests." *Reviews of Modern Physics* 85, 471 (2013).
8. Fein, Y. Y., Geyer, P., Zwick, P., et al. "Quantum superposition of molecules beyond 25 kDa." *Nature Physics* 15, 1242-1245 (2019).
9. White, H. G., Vera, J., Sylvester, A., & Dudzinski, L. "Emergent quantization from a dynamic vacuum." *Physical Review Research* 8, 013204 (2026). DOI: 10.1103/PhysRevResearch.8.013204